

including the P16, P14 and a compact desktop. Subsequent announcements have expanded on the software and Windows integration behind those machines. NVIDIA positions RTX Spark around creators, gamers and personal AI agents. Each audience places different demands on a PC. Creators need reliable acceleration in applications and enough memory for their projects. Gamers need compatibility, frame rates and responsive input. Local AI users need supported runtimes, model capacity and acceptable response times.

RTX Spark attempts to address all three through a common foundation: an Arm CPU, a Blackwell RTX GPU, unified memory and NVIDIA's software stack. The proposition extends beyond laptops to compact desktops, although the ProArt P16 remains the focus of today's review here.

Its rivals consequently come from several directions. Qualcomm's Snapdragon X2 Elite Extreme features a similar Windows on Arm system for comparison. Intel's Core Ultra 9 290HX Plus and AMD's Ryzen 9 9955HX represent powerful x86 laptop processors. AMD's Ryzen AI Max+ 395 is particularly relevant because it also combines substantial integrated graphics capability with a large shared memory architecture. We don't see the conventional Intel or AMD laptops with GeForce graphics getting dethroned anytime soon. However, it goes without saying that the two incumbent x86 companies have more to worry about with NVIDIA throwing its hat into the ring.

RTX Spark isn't the same as the DGX Spark. Sharing parts of NVIDIA's accelerated-computing ecosystem does not make them interchangeable products. The DGX Spark lives and beatles in the familiar Linux space where most development takes place, whereas with RTX Spark, the proposition is a Windows PC intended to run everyday applications, games, creative software and local AI.

Architecture

The N1X configuration in this ProArt uses ten Arm Cortex-X925 performance cores and ten Cortex-A725 efficiency cores. NVIDIA uses the Grace CPU branding and developed the CPU, I/O and memory tile with MediaTek. The core arrangement is distinct from both Qualcomm's Oryon implementation and the x86 designs from Intel and AMD. The performance cores handle demanding work, while the efficiency cores provide additional throughput and a lower-power option for suitable tasks. Actual behaviour depends on scheduling, power management and the app. 20 cores do not translate into a predictable advantage over an 18-core Qualcomm processor or a 16-core AMD CPU with simultaneous multithreading.

NVIDIA specifies maximum performance-core clocks of up to 4.0 – 4.1 GHz, with efficiency cores reaching 2.8 GHz. On the graphics side, the larger N1X configuration contains 48 streaming multiprocessors, packing in 6,144 CUDA cores, with fourth-generation RT cores and fifth-generation Tensor cores. Like always, CUDA cores execute general graphics and compute work, RT cores are used to accelerate ray-tracing operations, and Tensor cores handle supported AI calculations.

The Tensor hardware matters to both local AI and gaming. It supports the lower-precision calculations used by compatible models, as well as neural rendering technologies. NVIDIA advertises up to 1 petaflop of FP4 AI performance. However, that 1 petaflop figure should not be read as directly comparable with an NPU's INT8 TOPS rating or as a prediction of language-model output speed.

Specification	ASUS ProArt P16 H7607BA
Platform	NVIDIA RTX Spark N1X
CPU	20 cores: ten Cortex-X925 and ten Cortex-A725
Graphics	Blackwell RTX, 6,144 CUDA cores
Memory	64 GB unified memory
GPU memory allocation	32 GB
Storage	1 TB SSD
Display	16-inch OLED, 2880 × 1800, 16:10, approximately 212 ppi
Refresh rate	120 Hz
Operating system	Windows 11 Pro 26H1, build 28000.3086
NVIDIA driver	617.42
Installed local model	Qwen3.8-27B-UD-Q4_K_M, with Hermes
India price	Not available. Estimated at ₹1,99,999.